

University Administrative Copilot

Intent understanding, routing & structured service requests

Group 10 · NLP Course Project 2026 (AI University) · Week 1 Project Proposal · Roll Nos: J054, J055, J056, J057, J059

1. Problem	Students and faculty raise administrative requests in free text (emails, help-desk forms, chat groups, counter queues), while the procedures that answer them are scattered across the Student Resource Book, examination/attendance rules, fee and scholarship pages, hostel rules and circulars. People do not know which office handles what or which documents to bring ; staff answer the same questions repeatedly; requests get misrouted and re-forwarded, delaying time-sensitive items such as exam registration, fee deadlines and scholarship forms.
2. Users	Primary: students (UG/PG, hostel residents) and faculty raising requests. Secondary: administrative staff (exam cell, accounts/fees, scholarship cell, hostel office, student services) who receive pre-classified, pre-filled tickets. Programmatic: other AI University apps (portal chatbot, help-desk, Group 4 Knowledge Assistant, Group 6 Early-Warning) consuming our API.
3. Existing work	Commercial ticket triage (ServiceNow Virtual Agent, Zendesk/Freshdesk AI) and intent-slot frameworks (Rasa, Dialogflow); research benchmarks for intent detection incl. out-of-scope queries (CLINC150, BANKING77), joint intent-slot models (JointBERT) and few-shot classifiers (SetFit). Gap: generic FAQ bots are not grounded in institution-specific procedures and rarely emit a routed, structured service request.
4. Input / Output	Input: one free-text message + optional metadata (channel, requester_role), e.g. "I lost my ID card. What should I do?" Output (JSON): intent, department, entities, required documents, next action, cited sources, draft service request, confidence and a human-review flag. <pre>POST /route-request → {"intent":"id_card.replacement", "intent_confidence":0.93, "department":"Academic Administration", "entities":{"student_id":"<masked>", "document":"ID card", "date":"yesterday"}, "required_documents":[...], "next_action":"Submit replacement request", "sources":[{"doc":"student_resource_book.pdf", "section":"Identity Card"}], "service_request":{"type":"ID_CARD_REPLACEMENT", "fields":{"...}, "missing_fields":["program"]}, "needs_human_review":false}</pre>
5. NLP task	A pipeline of four NLP sub-tasks: (a) intent classification – 9 coarse intents from the brief (certificates, transcripts, ID cards, hostel, examinations, fees, scholarships, attendance, academic registration) + out_of_scope, refined into ~25 sub-intents (e.g. ID card → lost / damaged / correction); (b) NER / slot filling – STUDENT_ID, PROGRAM, SEMESTER, COURSE, DATE, AMOUNT, DOCUMENT_TYPE, HOSTEL_ROOM; (c) semantic retrieval of the governing procedure with section-level citations; (d) constrained generation of a schema-valid service request.
6. Dataset	• Knowledge corpus: ~30–50 public NMIMS/MPSTME documents (Student Resource Book, exam & attendance rules, fee/scholarship pages, hostel rules, academic calendar, circulars), chunked by section with metadata. • Query set (target ≥1,500 labelled): queries written by the team and classmates via an anonymous form (no PII), plus verified augmentation (paraphrases, typos, Hinglish code-mixing) used only in training; out-of-scope negatives incl. the CLINC150 OOS split. ~500 queries BIO-annotated for entities; ~100 procedure questions with gold source sections. • Split: 70/15/15 stratified by intent; test set = human-written queries only (no synthetic leakage). Synthetic IDs, public documents only, PII masked before any external LLM call.
7. Proposed approach	• Baseline: TF-IDF (word + char n-grams) + Logistic Regression / Linear SVM for intent; regex + gazetteer rules for entities; BM25 keyword retrieval; static intent→department table. • Improved: fine-tuned DistilBERT/MiniLM vs. SetFit sentence-transformer classifier with calibrated confidence and an out-of-scope threshold; transformer token-classification NER (+ rules for IDs/dates); hybrid BM25 + dense retrieval (bge-small / MiniLM, FAISS) with cross-encoder re-ranking; LLM used only to phrase answers from retrieved chunks with mandatory citations and to fill the JSON schema — routing and required documents come from deterministic mappings/retrieval, not free generation. • Evaluation: accuracy, macro-F1, confusion matrix and OOS recall (intent); entity-level P/R/F1 via seqeval (NER); routing accuracy; Recall@k and MRR (retrieval); answer correctness, citation accuracy and hallucination rate; slot-fill completeness; calibration (ECE). Baseline vs. improved on the same test set + 5–10 analysed failures. • Asset: Python package + FastAPI – POST /route-request, POST /extract-entities, GET /intents – with documented schema, README and example calls.
8. Institutional value	24×7 first response with policy-grounded, cited answers; fewer misrouted emails and tickets; less repetitive load on office staff; tickets arrive pre-filled with the right documents listed; demand analytics (e.g. transcript spikes before placement season) for planning. Integration: input screened by Group 12; circular summaries from Group 11 keep our corpus current; retrieval index shared with Group 4; distress-like messages handed off to Group 6. Responsible AI: advisory only – the copilot never approves or rejects a request; low-confidence or out-of-scope cases go to a human desk; answers without a source say “not found”.

9. Initial architecture — low-confidence or out-of-scope requests exit at the confidence gate to a human help-desk

